Digital Signal Processing – Laboratory

Spectral Analysis of Deterministic Signals

Task: Signal synthesis using the **IDFT in matrix notation**

In this notebook you will:

1. Define a spectrum vector $X[\mu]$ (given in the exercise variants).
2. Build the **index matrix** K and the **IDFT matrix** W.
3. Synthesize the discrete-time signal $x[n]$ using matrix multiplication for different values of N.
4. Plot the synthesized signal (real/imaginary parts and magnitude).

> **Notation**
> μ – frequency index (DFT bin)
> n – time index
> N – IDFT length (number of samples in one period)

In [17]:
```
import numpy as np
import matplotlib.pyplot as plt

# Make plots a bit nicer in Jupyter
plt.rcParams["figure.figsize"] = (9, 3.8)
plt.rcParams["axes.grid"] = True

np.set_printoptions(precision=3, suppress=True)
```

Theory: IDFT in matrix form

The IDFT is

[x[n] = \frac{1}{N} \sum_{\mu=0}^{N-1} X[\mu] , e^{j 2\pi \mu n/N}, \quad n=0,\dots,N-1.]

Define the **index matrix**

[K[n,\mu] = n\cdot \mu]

and the **IDFT matrix**

[W[n,\mu] = e^{j 2\pi K[n,\mu]/N}.]

Then the synthesis becomes a single matrix multiplication:

[\mathbf{x} = \frac{1}{N}, \mathbf{W}, \mathbf{X}.]

> In this lab, the given vectors are interpreted as **DFT/IDFT coefficients** $\mathbf{X}$.

Choose the variant (given spectrum vector $X[\mu]$)

Each variant defines a column vector $\mathbf{X}$ (real values in the assignment, but the IDFT result may be complex).

You can switch variants by changing `variant_id`.

In [18]:
```
# ----- Variant selection -----
variants = {
    1: np.array([6, 2, 4, 3, 4, 5, 0, 0, 0, 0], dtype=complex),
    2: np.array([10, 5, 6, 6, 2, 4, 3, 4, 5, 0, 0, 0, 0], dtype=complex),
    3: np.array([6, 2, 4, 3, 4, 5, 0, 0, 0, 0], dtype=complex),
    4: np.array([6, 2, 4, 3, 4, 5, 0, 0, 0], dtype=complex),
    5: np.array([6, 4, 4, 5, 3, 4, 5, 0, 0, 0, 0], dtype=complex),
}

variant_id = 1  # <-- change to 1..5

X = variants[variant_id].reshape(-1, 1)  # column vector (N0 x 1)
N0 = X.shape[0]

print(f"Variant {variant_id}: length N0 = {N0}")
print("X[μ] =")
print(X.T)
```

```
Variant 1: length N0 = 10
X[μ] =
[[6.+0.j 2.+0.j 4.+0.j 3.+0.j 4.+0.j 5.+0.j 0.+0.j 0.+0.j 0.+0.j 0.+0.j]]
```

Helper functions: build K, build W, and perform the IDFT

We will build matrices explicitly to match the assignment statement.

In [19]:
```
def build_K_W(N: int):
    """Return index matrix K and IDFT matrix W for a given length N."""
    n = np.arange(N).reshape(-1, 1)          # (N x 1)
    mu = np.arange(N).reshape(1, -1)        # (1 x N)
    K = n @ mu                               # (N x N) outer product => K[n, μ]
    W = np.exp(1j * 2 * np.pi * K / N)      # (N x N)
    return K, W

def idft_matrix(X: np.ndarray, N: int):
    """IDFT synthesis x = (1/N) * W * X, where X is (N x 1)."""
    K, W = build_K_W(N)
    x = (1 / N) * (W @ X)
    return x, K, W

def pad_or_trim_X(X: np.ndarray, N: int):
    """Return X resized to length N (zero-pad if needed, trim if too long)."""
    X = X.reshape(-1, 1)
    if X.shape[0] == N:
        return X
    if X.shape[0] < N:
        Xp = np.zeros((N, 1), dtype=complex)
        Xp[:X.shape[0], 0] = X[:, 0]
        return Xp
    # if longer, keep first N bins (simple trimming for the exercise)
    return X[:N, :]
```

Step 1: show matrices K and W for a *small* N

For large N the matrices are big, so we usually **display them for a small example** (e.g., $N = 8$) and then use bigger N for synthesis and plotting.

In [20]:
```
N_show = min(8, N0)  # display up to 8x8 to keep it readable

X_show = pad_or_trim_X(X, N_show)
x_show, K_show, W_show = idft_matrix(X_show, N_show)

print(f"Showing matrices for N = {N_show}\n")

print("K matrix (n·μ):")
print(K_show)

print("\nW matrix = exp(j*2π*K/N):")
print(W_show)

print("\nSynthesized x[n] (for N_show):")
print(x_show.T)
```

```
Showing matrices for N = 8

K matrix (n·μ):
[[ 0  0  0  0  0  0  0  0]
 [ 0  1  2  3  4  5  6  7]
 [ 0  2  4  6  8 10 12 14]
 [ 0  3  6  9 12 15 18 21]
 [ 0  4  8 12 16 20 24 28]
 [ 0  5 10 15 20 25 30 35]
 [ 0  6 12 18 24 30 36 42]
 [ 0  7 14 21 28 35 42 49]]

W matrix = exp(j*2π*K/N):
[[ 1.   +0.j     1.   +0.j     1.   +0.j     1.   +0.j     1.   +0.j
   1.   +0.j     1.   +0.j     1.   +0.j   ]
 [ 1.   +0.j     0.707+0.707j  0.   +1.j    -0.707+0.707j -1.   +0.j
  -0.707-0.707j -0.   -1.j     0.707-0.707j]
 [ 1.   +0.j     0.   +1.j    -1.   +0.j    -0.   -1.j     1.   -0.j
   0.   +1.j    -1.   +0.j    -0.   -1.j   ]
 [ 1.   +0.j    -0.707+0.707j -0.   -1.j     0.707+0.707j -1.   +0.j
   0.707-0.707j  0.   +1.j    -0.707-0.707j]
 [ 1.   +0.j    -1.   +0.j     1.   -0.j    -1.   +0.j     1.   -0.j
  -1.   +0.j     1.   -0.j    -1.   +0.j   ]
 [ 1.   +0.j    -0.707-0.707j  0.   +1.j     0.707-0.707j -1.   +0.j
   0.707+0.707j -0.   -1.j    -0.707+0.707j]
 [ 1.   +0.j    -0.   -1.j    -1.   +0.j     0.   +1.j     1.   -0.j
  -0.   -1.j    -1.   +0.j    -0.   +1.j   ]
 [ 1.   +0.j     0.707-0.707j -0.   -1.j    -0.707-0.707j -1.   +0.j
  -0.707+0.707j -0.   +1.j     0.707+0.707j]]

Synthesized x[n] (for N_show):
[[ 3.  +0.j  -0.28+0.5j  0.75+0.5j  0.78-0.5j  0.5 +0.j   0.78+0.5j
   0.75-0.5j -0.28-0.5j]]
```

Step 2: synthesize and plot for different values of N

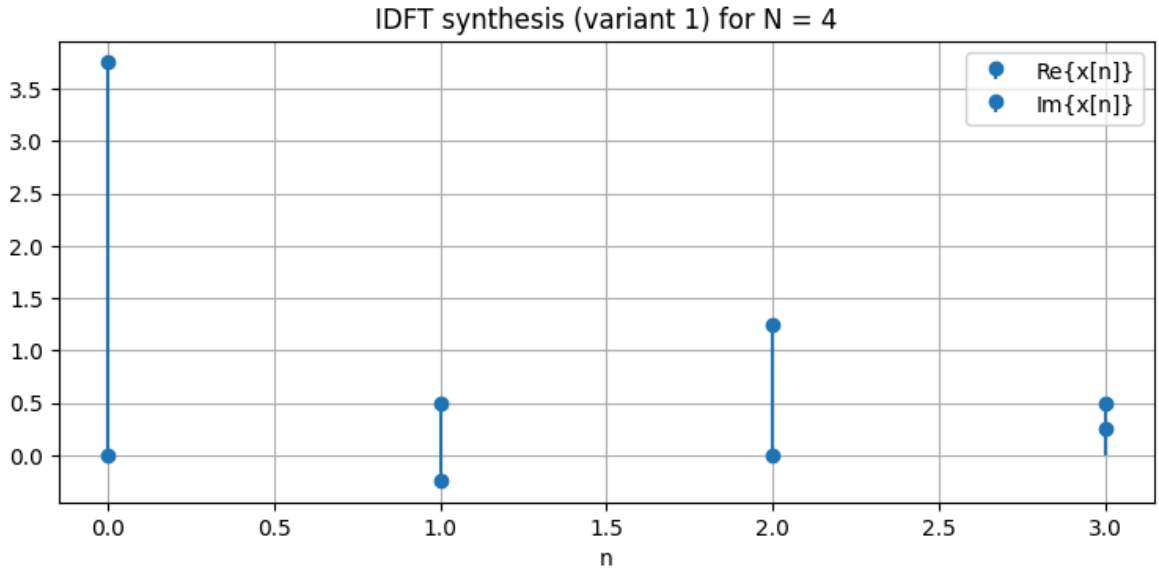
Idea: Changing N changes the assumed period and the frequency grid.
For this lab, we will keep the given coefficients and:

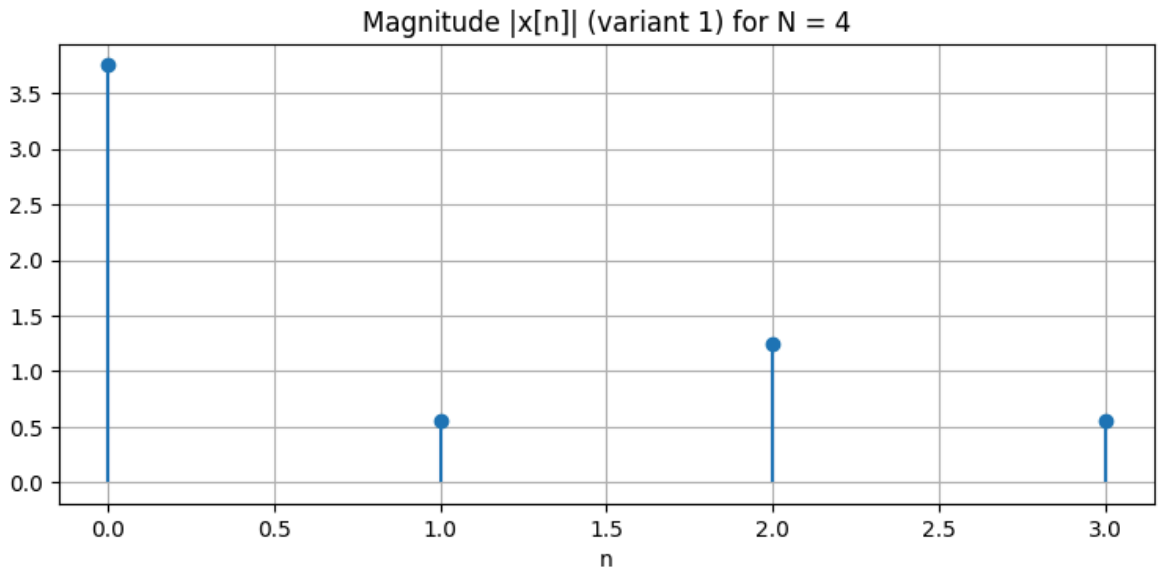
- Use $N = N_0$ (natural length of the provided vector),
- And a larger N (zero-padding the spectrum), e.g. $N = 2N_0$.

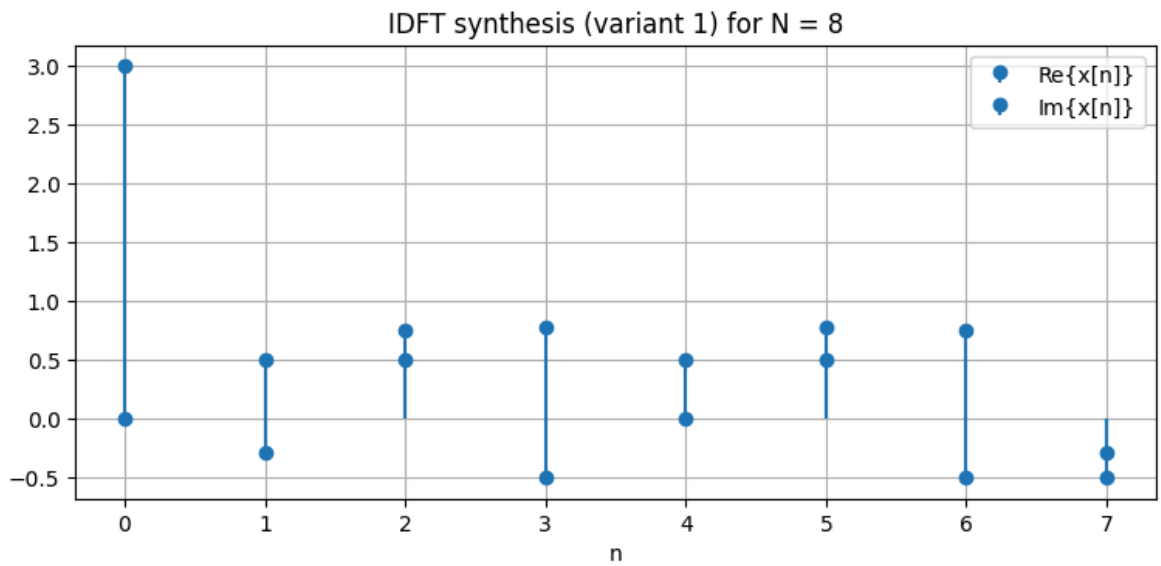
You can change `N_list` to any values you want (e.g., 16, 32, ...).

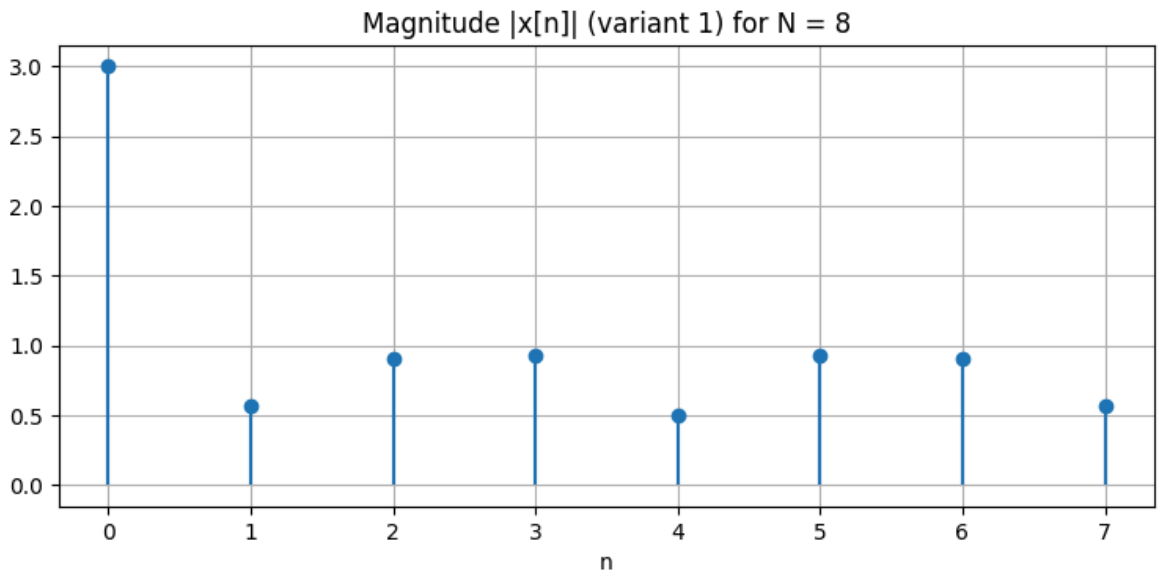
In [23]:
```python
import numpy as np
import matplotlib.pyplot as plt

N_list = [4, 4]  # instead of [N0, 2*N0]  # try also [N0, 16, 32] if you want

for N in N_list:
    XN = pad_or_trim_X(X, N)
    xN, K, W = idft_matrix(XN, N)

    n = np.arange(N)

    plt.figure()
```

```python
    # Real and imaginary parts
    plt.stem(n, np.real(xN[:, 0]), basefmt=" ", label="Re{x[n]}")
    plt.stem(n, np.imag(xN[:, 0]), basefmt=" ", label="Im{x[n]}")
    plt.title(f"IDFT synthesis (variant {variant_id}) for N = {N}")
    plt.xlabel("n")
    plt.legend()
    plt.show()

    plt.figure()
    # Magnitude
    plt.stem(n, np.abs(xN[:, 0]), basefmt=" ")
    plt.title(f"Magnitude |x[n]| (variant {variant_id}) for N = {N}")
    plt.xlabel("n")
    plt.show()
```









What to include in your report

1. The vector $\mathbf{X}$ for your assigned variant.
2. Matrices K and W (**show them for a small N so they fit on a page**, e.g., $N = 8$).
3. The synthesized signal $x[n]$ plotted for at least two values of N (e.g., $N = N_0$ and $N = 2N_0$).
4. A short comment: is $x[n]$ real or complex? how does increasing N affect the shape?